AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT131
Discrete Mathematics

Final Examination
Fall Semester 2025/2026

Date: 22 /Jan./2026

Number of Exam Pages: 3 pages (including this cover sheet)	**Time Allowed:** 2 hours

Instructions:

- Total Marks: **50**
- This exam consists of **5** questions.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Q–1: [5×2 marks] Choose the correct answer:

1.1: The negation of $(P \wedge Q) \rightarrow R$ is

a) $\neg(P \wedge Q) \rightarrow \neg R$
b) $\neg R \rightarrow \neg(P \wedge Q)$
c) $R \rightarrow (P \wedge Q)$
d) $P \wedge Q \wedge \neg R$
e) None of the above

1.2: Let A and B be sets such that $|A| = 4$ and $|B| = 3$. Which of the following is always true?

a) $|A \cap B| = 0$
b) $|A \cup B| = 7$
c) $|A \times B| = 12$
d) $|A - B| = 1$
e) None of the above

1.3: The hexadecimal representation of the octal number (1234) is

a) A70
b) 29C
c) 27A
d) A72
e) None of the above

1.4: Assume that R is a relation on $\{1, 2, 3, 4\}$ defined as follows: aRb if and only if $a + b$ is an odd number, then $R =$

a) $\{(1,2), (1,4), (2,3), (3,4)\}$
b) $\{(2,1), (4,1), (3,2), (4,3)\}$
c) $\{(1,2), (3,4)\}$
d) $\{(1,2), (2,1), (1,4), (4,1), (2,3), (3,2), (3,4), (4,3)\}$
e) None of the above

1.5: A graph with incident matrix of size 5×6 has number of edges equal

a) 30
b) 11
c) 5
d) 6
e) None of the above

Each question is worth 10 marks. Answer the following questions:

Q–2:

a) **[4 marks]** Using the truth table, determine whether the statements $\neg(p \oplus q) \vee (p \rightarrow r)$, and $p \rightarrow (q \vee r)$ are logically equivalent.

b) **[2+2 marks]** Determine whether each of the following quantifier is True or False, Justify your answer:

i. The graph $G = (\{a, b, c\}, \{(a, a), (a, b), (a, c)\})$ is a tree.
ii. $\exists x(3x = x \; div 3)$, domain is the set of integers.

c) **[1+1 mark each]** Let $a = 825$ and $b = 2^2 \times 11 \times 13$.

i. Find the prime factorization of a
ii. Are a and b relatively prime? Justify your answer.

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Q–3:

a) [2+1+1 marks] How many four character password can be made using only A, B, C, D, E, F, 1, 2, 3, 4, 5, 6,7,8 if

i. No character is reused.
ii. Characters can be repeated.
iii. Character can be repeated as long as they are not adjacent.

b) [2+2+2 marks] Of 10 girls in a class, 3 have blue eyes. If two of the girls are chosen at random, what is the probability that:

i. Both have blue eyes.
ii. Neither has blue eyes.
iii. At least one has blue eyes.

Q–4:

a) [2+2 marks] Let $R = \{(x, y), (y, y), (y, z), (z, y)\}$ be a relation on $A = \{x, y, z, w\}$.

i. Find the reflexive closure of R
ii. Find the symmetric closure of R

b) [2+2 marks] Let R, and S be relations on the set $\{a, b, c\}$ represented by the matrices below (where the rows and columns correspond to the letters listed in alphabetical order).

$$M_R = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \quad M_S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

Find the matrices that represent the relations:

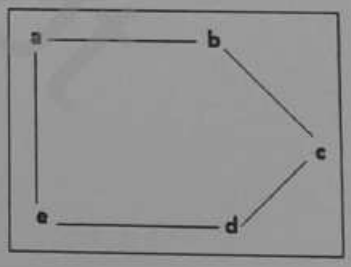
(i) $R \cup S$

(ii) $R \cap S$

c) [2 marks] Let $A = \{1, 3, 5, 10, 15, 20\}$, Draw the Hasse diagrams for the following poset $(A, |)$

Q–5:

a) [3 +3 marks] Given a graph $G = (V, E)$ with vertices a, b, c, d, e :



i. Write the adjacency matrix of G.
ii. Write the incident matrix of G.

b) [4 marks] up a binary tree to sort the following list of numbers:

$$20, 30, 10, 15, 35, 13, 0, 32, 27.$$

End of questions

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